MATH 301: Homework 2

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Section 2.3: q4

Let $S_4 := \left\{1 - \frac{(-1)^n}{n} : n \in \mathbb{N}\right\}$. *Find* $\inf S_4$ *and* $\sup S_4$.

Let $f(n) := 1 - \frac{(-1)^n}{n}$.

Consider the first few terms in the sequence:

$$\begin{aligned}
f(1) &= 1 - \frac{(-1)^1}{1} = 1 + 1 = 2 \\
f(2) &= 1 - \frac{(-1)^2}{2} = 1 - \frac{1}{2} = \frac{1}{2} \\
f(3) &= 1 - \frac{(-1)^3}{3} = 1 + \frac{1}{3} = \frac{4}{3} \\
f(4) &= 1 - \frac{(-1)^4}{4} = 1 - \frac{1}{4} = \frac{3}{4} \\
&\dots
\end{aligned}$$

We can reconstruct this sequence into two subsequences based on whether the index is odd or even. Let K be the set of odd natural numbers and J be the set of even natural numbers:

$$K := \{k \in \mathbb{N} : 2 \nmid k\} \quad J := \{j \in \mathbb{N} : 2 \mid j\}$$

For odd $k \in K$, $(-1)^k = -1$, giving the function $g(k)$:

$$g(k) = 1 + \frac{1}{k}$$

For even $j \in J$, $(-1)^j = 1$, giving the function $h(j)$:

$$h(j) = 1 - \frac{1}{j}$$

Consider the first terms of $g(k)$ where $k \in \{1, 3, 5, \dots\}$:

$$g(1) = 2, \quad g(3) = \frac{4}{3}, \quad g(5) = \frac{6}{5}, \quad \dots$$

Observe that $g(k)$ is strictly decreasing. For $x \in K$:

$$\begin{aligned}
& x < x + 2 \\
\implies & \frac{1}{x} > \frac{1}{x+2} \\
\implies & 1 + \frac{1}{x} > 1 + \frac{1}{x+2} \\
\implies & g(x) > g(x+2)
\end{aligned}$$

Consider the first terms of $h(j)$ where $j \in \{2, 4, 6, \dots\}$:

$$h(2) = \frac{1}{2}, \quad h(4) = \frac{3}{4}, \quad h(6) = \frac{5}{6}, \quad \dots$$

Observe that $h(j)$ is strictly increasing. For $z \in J$:

$$
\begin{aligned}
z &< z+2 \\
\implies \frac{1}{z} &> \frac{1}{z+2} \\
\implies -\frac{1}{z} &< -\frac{1}{z+2} \\
\implies 1-\frac{1}{z} &< 1-\frac{1}{z+2} \\
\implies h(z) &< h(z+2)
\end{aligned}
$$

Because $g(k)$ is strictly decreasing, its supremum is its first term (2). Because $h(j)$ is strictly increasing, its infimum is its first term ($\frac{1}{2}$). Therefore, for the entire set S_4:

$$\therefore \sup S_4 = 2, \quad \inf S_4 = \frac{1}{2}$$

Section 2.3: q5

Find the infimum and supremum, if they exist, of each of the following sets.

(a) $A := \{x \in \mathbb{R} : 2x+5 > 0\}$, *(b)* $B := \{x \in \mathbb{R} : x+2 \geq x^2\}$,
(c) $C := \{x \in \mathbb{R} : x < 1/x\}$, *(d)* $D := \{x \in \mathbb{R} : x^2 - 2x - 5 < 0\}$.

Part (a): $A := \{x \in \mathbb{R} : 2x+5 > 0\} \implies \inf A = \frac{-5}{2}, \ \nexists \ \sup A$
Part (b): $B := \{x \in \mathbb{R} : x+2 \geq x^2\} \implies \inf B = -1, \ \sup B = 2$
Part (c): $C := \{x \in \mathbb{R} : x < 1/x\} \implies \nexists \ \inf B, \ \sup C = 1$
Part (d): $D := \{x \in \mathbb{R} : x^2 - 2x - 5 < 0\} \implies \inf D = 1 - \sqrt{6}, \ \sup D = 1 + \sqrt{6}$

Section 2.3: q6

Let $S \subseteq \mathbb{R}, S \neq \emptyset$ *and bounded below. Prove that* $\inf S = -\sup\{-s : s \in S\}$

S is bounded below $\implies \exists \ u$ s.t. $u := \inf S$

$u := \inf S \implies \forall \ s \in S, \ (u \leq s) \wedge (u$ is the largest lowerbound$)$

u is the largest lowerbound $\implies$ If w is a lowerbound. Then $w \leq u$

Let $|S| = n \in \mathbb{N}$ Consider ordering of $S := \{s_1 \leq s_2 \leq ... \leq s_n\}$

Define $T := \{-s : s \in S\}$ Observe $u := \inf S \implies u \leq s_1$

Consider ordering of $T := \{-s_1 \geq -s_2 \geq ... \geq -s_n\}$

S is bounded below $\implies -S$ is bounded above $\implies \exists \ \sup -S$

Observe $u := \inf S \implies -u \geq -s_1$

Thus $-u \geq -s_1 \geq -s_2 \geq ... \geq -s_n \implies \forall \ x \in -S, -u \geq x \implies -u$ is an upperbound of S

Assume $\exists \ (z < u)$ s.t. $\forall \ x \in -S, z \geq x \qquad z := (-u + \frac{1}{n}), \ n \in \mathbb{N}$

$$\implies -u + \frac{1}{n} > -s_1 \implies u - \frac{1}{n} < s_1$$

Section 2.3: q8

Let $S \subseteq \mathbb{R}, S \neq \emptyset$. Show $u \in \mathbb{R}$ is an upper bound of $S \iff (t \in \mathbb{R}) \wedge (t > u) \implies (t \notin S)$

Prove: $u \in \mathbb{R}$ *is an upper bound of* $S \implies ((t \in \mathbb{R}) \wedge (t > u) \implies (t \notin S))$

$$\text{Suppose for the sake of contracdiction: } (t \in S) \wedge (t \in \mathbb{R}) \wedge (t > u)$$

$$u \in \mathbb{R} \text{ is an upper bound of } S \implies \forall\ s \in S, u \geq s$$

$$(t \in S) \wedge (t > u) \implies u \text{ is not an upperbound of } S \ \lightning$$

$$\therefore\ (u \in \mathbb{R} \text{ is an upper bound of } S) \implies ((t \in \mathbb{R}) \wedge (t > u) \implies (t \notin S))$$

Prove: $((t \in \mathbb{R}) \wedge (t > u) \implies (t \notin S)) \implies$ *($u \in \mathbb{R}$ is an upper bound of S)*

Consider the contrapositive:

$$(u \in \mathbb{R} \text{ is NOT an upper bound of } S) \implies \exists\ t \text{ s.t. } ((t \in \mathbb{R}) \wedge (t > u) \wedge (t \in S))$$

If $u \in \mathbb{R}$ is NOT an upper bound of $S \implies \exists\, t \in (S \cup \mathbb{R}) \ni t > u \implies ((t \in \mathbb{R}) \wedge (t > u) \wedge (t \in S))$

Section 2.3: q10

Show that if A and B are bounded subsets or $\mathbb{R}$. Then $A \cup B$ is a bounded set.
Show that $\sup(A \cup B) = \sup\{\sup A, \sup B\}$ "

$$\text{Let} \quad u := \sup S, \quad v := \sup B, \quad w := \sup\{u, v\}$$

$$(x \in A \implies x \leq u \leq w) \wedge (x \in B \implies x \leq v \leq w) \implies w \text{ is an upper bound of } A \cup B$$

$$z := \text{ any upper bound of } A \cup B \implies z \text{ is an upper bound of } A \text{ and } B$$

$$\implies u \leq z \wedge v \leq z \implies w \leq z$$

$$\therefore w = \sup(A \cup B)$$

Section 2.4: q2

If $S := \{\frac{1}{n} - \frac{1}{m} | n, m \in \mathbb{N}\}$ find $\inf S$ and $\sup S$

Consider for the following:

$$n \in \mathbb{N} \implies 0 < \frac{1}{n} \leq 1$$

$$m \in \mathbb{N} \implies -1 < -\frac{1}{m} \leq 1$$

Finding Supremum

Find the maximum value of $\frac{1}{n}$ and minimum $\frac{1}{m}$.

$$\text{Considering the bounds: } \begin{cases} \implies \dfrac{1}{n} = 1 \\ \implies -\dfrac{1}{m} = 0 \end{cases}$$

Note $\sup S = 1$ this statement holds if 1 the smallest upperbound of S.

Consider: $\lim_{m\to\infty}(1 - \frac{1}{m})$

$$|(1 - \frac{1}{m}) - 1| < \epsilon \iff |-\frac{1}{m} + (1-1)| < \epsilon \iff |-\frac{1}{m}| < \epsilon \iff |\frac{1}{m}| < \epsilon \iff \frac{|1|}{m} < \epsilon$$

$$\leftarrow m > \frac{1}{\epsilon} \leftarrow \text{Archimedian Property} \leftarrow \frac{1}{\epsilon} \in \mathbb{R}$$

$$\therefore \lim_{m\to\infty}(1 - \frac{1}{m}) = -1$$

Consider: $\lim_{n\to\infty}(\frac{1}{n} - 1)$

$$|(\frac{1}{n} - 1) + 1| < \epsilon \iff |\frac{1}{n} + (1-1)| < \epsilon \iff |\frac{1}{n}| < \epsilon \iff \frac{|1|}{n} < \epsilon \iff n > \frac{1}{\epsilon}$$

$$\leftarrow \text{Archimedian Property} \leftarrow \frac{1}{\epsilon} \in \mathbb{R}$$

$$\therefore \lim_{n\to\infty}(\frac{1}{n} - 1) = 1$$

Thus, we can say that the $\sup S = 1$.

Finding Infinimum

Find the minimum value of $\frac{1}{n}$ and maximum $\frac{1}{m}$.

$$\text{Considering the bounds: } \begin{cases} \implies \dfrac{1}{n} = 0 \\ \implies -\dfrac{1}{m} = -1 \end{cases}$$

Consider: $\lim_{n\to\infty}(\frac{1}{n} - 1)$

Section 2.4: q7

Let A, B be bounded nonempty subsests of $\mathbb{R}$, and let $A + B := \{a + b | a \in A, b \in B\}$.
Prove $\sup(A + B) = \sup A + \sup B$ and $\inf(A + B) = \inf A + \inf B$.

Consider for the following:

$$\sup A = a_s \implies \forall\ a \in A, a < a_s$$
$$\sup B = b_s \implies \forall\ b \in B, b < b_s$$

Observe:

$$\begin{array}{r} (a < a_s) \\ +\ (b < b_s) \\ \hline a + b < a_s + b_s \end{array}$$

Observe, $a + b$ represent arbitrary elements of $A + B$
Therefore, $\sup A + B = a_s + b_s$

$$\inf A = a_i \implies \forall\ a \in A, a > a_i$$
$$\inf B = b_i \implies \forall\ b \in B, b > b_i$$

Observe:

$$\begin{array}{r} (a > a_i) \\ +\ (b > b_i) \\ \hline a + b > a_i + b_i \end{array}$$

Observe, $a + b$ represent arbitrary elements of $A + B$
Therefore, $\inf A + B = a_i + b_i$

Section 2.4: q9

> *Let $X = Y := \{x \in \mathbb{R} : 0 < x < 1\}$. Define $h : X \times Y \to \mathbb{R}$ by $h(x, y) := 2x + y$.*
>
> *(a) For each $x \in X$, find $f(x) := \sup\{h(x, y) : y \in Y\}$; then find $\inf\{f(x) : x \in X\}$.*
>
> *(b) For each $y \in Y$, find $g(y) := \inf\{h(x, y) : x \in X\}$; then find $\sup\{g(y) : y \in Y\}$. Compare with the result found in part (a).*

Item (a)

$$\forall\ x \in X, f(x) := \sup\{h(x, y) : y \in (0, 1)\} \implies f(x) = 2x + 1$$
$$f(x) = 2x + 1 \wedge x \in (0, 1) \implies \inf\{h(x, y) : x \in X\} = 2(0) + 1 = 1$$

Item (b)

$$\forall\ y \in Y, g(i) := \inf\{h(x, y) : x \in (0, 1)\} \implies g(y) = 2(0) + y = y$$
$$g(y) = y \wedge y \in (0, 1) \implies \sup\{g(y) : y \in Y\} = 1$$

Section 2.4: q14

> *If $y > 0$ show $\exists\ n \in \mathbb{N} \ni \frac{1}{2^n} < y$*

Recall:

$$a > b \implies \frac{1}{a} < \frac{1}{b}$$

Observe:

$$\begin{aligned}
&P(n) : n < 2^n \\
&P(1) : 1 < 2^1 \\
&P(k) : k < 2^k \\
&2k < 2^{k+1} \\
&2k + 1 < 2k < 2^{k+1} \\
&\therefore P(n) \text{ holds } \forall\ n \in \mathbb{N}
\end{aligned}$$

Thus we can say $\frac{1}{2^n} < \frac{1}{n}$. By Archimedian Property as $y > 0 \implies \frac{1}{n} \leq y$

$$\frac{1}{2^n} < \frac{1}{n} \leq y$$

Section 2.4: q15

Modify the argument in Theorem 2.4.7 to show that $\exists\ y > 0 \in \mathbb{R}$ s.t. $y^2 = 3$

> **Theorem 2.4.7** $\exists\ x > 0 \in \mathbb{R} \ni x^2 = 2$

$$S = \{s \in \mathbb{R} \mid s \geq 0 \text{ and } s^2 < 3\}$$

Existence of Supremum

$$0^2 = 0 < 3 \implies 0 \in S \implies S \neq \emptyset$$

$$\text{Let } \forall\ s \in S, s < 3 \text{ Suppose } \exists\ s \in S \ni s \geq 3 \implies s^2 \geq 9 > 3 \;\lightning \implies \text{S is bounded above.}$$

$$S \neq \emptyset \wedge S \text{ is bounded above } \implies \exists\ x \in \mathbb{R} \ni x = \sup S \text{ (By Completeness Property.)}$$

Trichotomy

By Trichotomy, one and only one, of the following must be true:

$$(x^2 < 3) \qquad (x^2 > 3) \qquad (x^2 = 3)$$

(Case: $(x^2 > 3)$**)**

Suppose for the sake of contracdiction x is the least upper bound of S

$$\text{Show: } \forall\ n \in \mathbb{N}, (x - \frac{1}{n})^2 > 3 \implies \forall\ s \in S, (x - \frac{1}{n})^2 > s^2 \implies (x - \frac{1}{n}) > s$$

Observe the following:

$$(x - \frac{1}{n})^2 = x^2 - \frac{2x}{n} + \frac{1}{n^2}$$

$$x^2 - \frac{2x}{n} + \frac{1}{n^2} \xrightarrow{\textit{Dropping } \frac{1}{n^2}} (x - \frac{1}{n})^2 > x^2 - \frac{2x}{n}$$

To hold supposition, the inequality must be stricly greater than three.

$$x^2 - \frac{2x}{n} > 3$$
$$x^2 - 3 > \frac{2x}{n}$$
$$\frac{1}{x^2-3} < \frac{n}{2x}$$
$$\frac{2x}{x^2-3} < n$$

Since $x^2 > 3 \implies (x^2 - 3) > 0 \implies \exists\ n \ni x > \frac{1}{n} > 0$.

$$(x - \frac{1}{2})^2 > 3 \implies \forall\ s \in S, s^2 < 3 < (x - \frac{1}{n})^2$$
$$\implies x - \frac{1}{n} \text{ is an upper bound of } S$$
$$x - \frac{1}{n} < x \implies x \text{ is not the least upper bound of } S \ ↯$$
$$\therefore\ (x^2 \not> 3)$$

(Case: $(x^2 < 3)$**)**

Observe the following:

$$(x + \frac{1}{n})^2 = x^2 + \frac{2x}{n} + \frac{1}{n^2}$$

Note $\forall\ n \in \mathbb{N}, \frac{1}{n^2} \leq \frac{1}{n}$

$$x^2 + \frac{2x}{n} + \frac{1}{n^2} \leq x^2 + \frac{2x+1}{n}$$
$$\implies x^2 + \frac{2x+1}{n} < 3$$
$$\implies \frac{2x+1}{n} < 3 - x^2$$
$$\implies \frac{n}{2x+1} > \frac{1}{3-x^2}$$
$$\implies n > \frac{2x+1}{3-x^2}$$

As $x^2 < 3 \implies 3 - x^2 > 0$. Therefore by Archimedian Property $\implies (x + \frac{1}{n}) \in S$ $\implies x$ is not an upperbound ↯.

$$\therefore\ (x^2 \not< 3)$$

As $x^2 \not< 3$ ***and*** $x^2 \not> 3$***. By Trichotomy*** $x^2 = 3$